

A Hybrid Approach to Driver Fatigue Detection on Edge Devices: Fusion of CNN and Landmark-Based Features

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Abstract. This research paper describes a robust, hybrid system for fatigue detection aimed for real-time driver monitoring on resource-limited edge devices. The system integrates lightweight Convolutional Neural Network (CNN) models with landmark-based geometric analysis for accuracy and computational efficiency enhancement. The system utilizes two distinct CNNs for eye state detection and yawning detection, which run in parallel with geometric feature extraction to compute inputs such as the EAR (Eye Aspect Ratio) and MAR (Mouth Aspect Ratio). A hybrid decision-making reasoning integrates the inputs of both the deep learning and geometric methods to offer more robust predictions of fatigue. Experimental results validate the system's high accuracy, with the eye state detection model scoring 98.34%, and the yawn detection model scoring 96.84%. This hybrid method offers a practical and efficient solution to enhancing road safety through intelligent driver assistance systems deployable on embedded hardware.

Keywords: fatigue · CNN · computer vision · edge computing.

1 Introduction

Fatigue-induced impairments remain a major cause of road accidents, making driver monitoring systems (DMS) essential for transportation safety. Traditional approaches, such as physiological monitoring (EEG, ECG), achieve high accuracy but are intrusive and impractical for daily driving, while vision-based methods using facial tracking often require high computational resources and are sensitive to occlusions and environmental variability. To address these challenges, this study presents a hybrid fatigue detection framework combining CNN-based

eye and yawn classification with real-time landmark analysis. The integration of geometric features (EAR, MAR, head tilt) and occlusion handling enhances robustness, while lightweight CNN models enable deployment on edge platforms like Raspberry Pi. This fusion approach improves detection accuracy, reduces false positives, and ensures efficient real-time performance, making it a practical solution for intelligent driver assistance systems¹

2 Objectives

The objective of this research is to design a real-time driver fatigue detection framework capable of identifying early warning signs, thereby enabling timely intervention to enhance road safety. The system adopts a hybrid approach that combines deep learning-based visual analysis with geometric feature-based positioning, ensuring robustness and computational efficiency suitable for embedded platforms. The key objectives are:

- Develop and train lightweight convolutional neural network (CNN) models for detecting fatigue-related visual cues such as eye state and yawning.
- Integrate spatial and landmark-based analysis to improve robustness against occlusions and environmental variability.
- Optimize the framework for deployment on edge devices (e.g., Raspberry Pi, low-power microprocessors) with reliable, low-latency inference.
- Evaluate system performance in terms of detection accuracy, real-time responsiveness, and feasibility for real-world vehicular deployment.

By fusing model-driven inference with geometry-aware analysis in a resource-efficient framework, the system aims to deliver a practical and deployable solution for intelligent driver monitoring on edge hardware.

3 Literature Survey

Fatigue detection systems are generally classified into three categories: physiological signal-based, vehicle behavior-based, and vision-based. Physiological approaches such as EEG, ECG, or heart rate monitoring provide high accuracy but are invasive and impractical for real-world driving due to the reliance on wearable sensors. Vehicle behavior-based methods, including lane drift or steering wheel analysis, are less intrusive but often unreliable under changing traffic or environmental conditions.

Vision-based approaches have gained prominence for their non-invasive, cost-effective, and real-time detection capabilities using standard cameras. These systems typically rely on cues such as eye blinking, prolonged eyelid closure, yawning, or head nodding. To address challenges like occlusion and illumination variability, recent studies increasingly combine deep learning with geometric or signal processing techniques.

For instance, Savaş and Becerikli [1] introduced a multi-task CNN that jointly evaluates eye and mouth regions for fatigue classification, while Zhuang et al.

[2] improved eye state detection by integrating segmentation for iris and pupil localization. Du et al. [3] advanced the field with a multimodal RNN that fuses facial features with physiological signals like heart rate, offering more robust detection in complex driving scenarios. Complementing these, Khan et al. [4] applied wavelet-based entropy features to low-resolution videos, demonstrating resilience in constrained environments.

Overall, these works illustrate a clear shift toward lightweight, hybrid fatigue detection systems optimized for edge deployment, combining deep learning accuracy with geometric feature robustness. A comparative overview of representative studies is provided in Table 1.

4 System Architecture

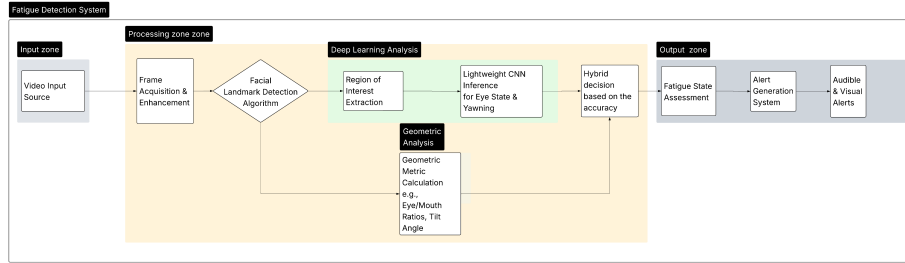


Fig. 1. Block diagram of the proposed system.

The driver fatigue detection system is based on a real-time, hybrid architecture that combines geometric facial analysis with deep learning inference. This two-methodology design provides strong and reliable monitoring of a driver’s cognitive state by establishing redundancy across two different analytical paths. The architecture is optimized for deployment on embedded computing platforms with a focus on computational efficiency and real-time response. The flow of operations of the system is organized into five main steps: (1) Preprocessing and Data Acquisition, (2) Facial Landmark Localization, (3) Multi-Modal Feature Extraction, (4) Hybrid Decision Logic, and (5) Alert Generation.

4.1 Data Acquisition and Preprocessing

The fatigue detection pipeline initiates with the acquisition of a video stream from an onboard camera module. Each captured frame undergoes preprocessing, including CLAHE (Contrast Limited Adaptive Histogram Equalization), to enhance local contrast and improve visibility of facial features. This step is crucial for maintaining system reliability under varying lighting conditions commonly encountered in real-world driving environments.

Table 1. Summary of Key Driver Fatigue Detection Papers

Sr.	Author(s)	Work done	ML Algorithm	Classification Level	Methodology / Features Used
1	Savas et al. [1]	Multi-task CNN for fatigue detection	CNN	3-level fatigue	Eye closure, yawning (PERCLOS, FOM), NTHU-DDD and YawDD datasets
2	Zhang et al. [2]	Eye state detection using IR video	CNN	Binary	Eye openness and blink frequency using infrared face video
3	Wang et al. [3]	EEG-based fatigue detection	PSD + Entropy	Continuous	EEG theta, alpha, beta wave features from dry wearable EEG headset
4	Du et al. [4]	Vision + HR fusion for fatigue detection	MFRNN (CNN+RNN+Fuzzy)	Binary	Multimodal fusion of HR, eye and mouth openness using RGB-D camera
5	Zhuang et al. [5]	Iris-pupil based eye status model	CNN + Segmentation	Binary	Lightweight U-Net to segment pupil and iris for eye openness calculation
6	Khan et al. [6]	Transform-domain facial fatigue detector	DWT + DCT + Entropy	Multi-class (7)	Wavelet and entropy features from facial regions in low-res videos
7	Ma et al. [7]	Night-time depth video-based detection	Two-Stream 3D CNN	Binary	Temporal depth motion analysis from night-vision camera
8	Huang et al. [8]	FDCN (Fatigue Detection CNN)	CNN	Binary	Fatigue-specific CNN model using CEW and ZJU datasets with eye-mouth regions
9	Reddy and Swathi [9]	Mouth segmentation using thresholding	Threshold Method	Binary	Morphological segmentation of yawn area on YawDD dataset
10	Liu et al. [10]	Blink classification model	RBF Neural Net	Binary	Eye closure duration and blink frequency on simulated data
11	Coetzer and Hancke [11]	Classifier comparison	SVM, ANN, AdaBoost	Binary	Feature-based eye shape and motion evaluation with multiple classifiers
12	Luo et al. [12]	Lighting-robust eye detection	AdaBoost + tracking	Binary	Eye detection with tracking and lighting adaptation under real conditions
13	Ji et al. [13]	Multi-index fatigue fusion	CNN + Fusion	Multi-level score	Weighted score combining eye, mouth, and head pose inputs
14	Lyu et al. [14]	Random Forest fatigue classifier	Random Forest	Binary	Face posture and gesture features from NTHU-DDD and F-DDD datasets
15	Gwak et al. [15]	Mild drowsiness classification	SVM + LR + RF	Multi-class	32 physiological + behavioral features from driving simulator
16	Junaedi and Akbar [16]	Blink detection via iris tracking	Iris Tracker	Binary	PERCLOS from iris region tracked in frontal video streams
17	Miah et al. [17]	Eye closure timing estimation	Threshold Logic	Binary	Duration-based analysis of eye closure from video frame sequences
18	Xing et al. [18]	CNN fatigue classifier	CNN	Binary	Eye openness and face region classification using CNN model
19	Ursulescu et al. [19]	Rule-based eye blink detector	Rule-based	Binary	Blink detection using Viola-Jones + heuristics on webcam feed
20	Ji et al. [20]	CNN with decision layer	CNN + Decision Layer	Binary	Fusion of facial zone features and CNN classification logic

4.2 Facial Landmark Localization

A high-precision facial landmark detection algorithm is employed to localize key facial points in real-time. These landmarks provide the geometric basis for further feature extraction and are configured to detect only a single subject per frame, optimizing computational efficiency and avoiding false detections.

4.3 Multi-Modal Feature Extraction

To ensure robustness, the system extracts features through two parallel analysis paths:

Geometric Feature Analysis This path computes fatigue-related metrics directly from facial landmarks:

- **Eye Aspect Ratio (EAR):** Detects blinks and eye closure based on the ratio of vertical to horizontal distances of eye landmarks.
- **Mouth Aspect Ratio (MAR):** Measures mouth opening to detect yawns.
- **Head Pose Estimation:** Captures orientation and tilt of the driver’s head to identify potential drowsiness-related postures.

Deep Learning Feature Analysis The second path utilizes lightweight Convolutional Neural Networks (CNNs) for inference:

- **ROI Extraction:** Eye and mouth regions are dynamically extracted using landmark coordinates.
- **CNN Inference:** Cropped ROIs are passed through pre-trained CNN models optimized for edge deployment. The models output probabilistic scores for eye closure and yawning detection.

4.4 Hybrid Decision Logic

This module fuses the outputs from both geometric and deep learning paths to make robust fatigue predictions. Temporal State Buffering Recent prediction states are stored in a sliding window buffer to smooth output and prevent false positives due to transient movements. Fused Condition Logic Final fatigue classification is derived using logical operations: a fatigue condition is confirmed if either EAR falls below a threshold or the CNN reports a closed eye with high confidence. This hybrid logic ensures the detection of both fast (blinks) and prolonged (microsleep) fatigue cues. Drowsiness Determination The system flags a drowsiness event when sustained eye closure or repeated yawning is observed beyond clinically established thresholds.

4.5 Alert Generation

Upon detection of fatigue, the system initiates an alert mechanism. Alert Prioritization Detected events are categorized based on severity (e.g., Yawn, Head Tilt, Drowsiness, Severe Drowsiness). Auditory Feedback Alerts are issued through a buzzer or speaker. Each level of fatigue is associated with a distinct sound pattern or voice prompt, ensuring clear communication to the driver.

5 Evaluation Metrics

5.1 Detection Accuracy

The detection capabilities of the CNN-based and geometric classifiers were evaluated using standard classification metrics derived from the confusion matrix (True Positives, False Positives, True Negatives, and False Negatives). The metrics are defined as follows:

- **Precision:** Indicates the proportion of correctly identified fatigue events among all positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity):** Measures the system’s ability to correctly detect all actual fatigue instances, which is critical for ensuring driver safety.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score:** Provides a balanced metric between precision and recall, useful when the class distribution is imbalanced.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

5.2 Real-Time Performance

This category evaluates the system’s suitability for embedded deployment and its responsiveness under live camera input. Frames Per Second (FPS) Represents how many frames the system can process per second. A higher FPS implies smoother real-time performance. End-to-End Latency Time taken from frame capture to alert generation. Lower latency ensures faster driver feedback and better intervention timing. Resource Utilization Monitored using system logs, this includes CPU and memory usage to ensure sustainable performance on resource-limited edge devices like Raspberry Pi.

5.3 System Robustness

Robustness metrics ensure that the system operates reliably under challenging real-world conditions. The key metrics are:

- **Environmental Resilience:** Evaluated under diverse lighting conditions, presence of accessories (e.g., glasses), and partial occlusions to validate generalization and stability.
- **False Alarm Rate (FAR):** Represents the proportion of non-fatigued frames incorrectly classified as fatigue events.

$$\text{FAR} = \frac{FP}{FP + TN}$$

- **Missed Detection Rate (MDR):** Represents the proportion of actual fatigue events that were not detected.

$$\text{MDR} = \frac{FN}{TP + FN}$$

6 Proposed Methodology

The proposed fatigue detection framework is based on a dual-pathway hybrid architecture designed for real-time operation on edge hardware. It fuses deep learning-based CNN classification with landmark-driven geometric analysis, supported by a hybrid logic engine for robust fatigue detection.

6.1 CNN Model Design

Two independent CNN models are designed—one for detecting eye states and another for yawning. Both share a 15-layer deep architecture with 256 neurons per layer, achieving high accuracy (99.96%) in offline training. They are deployed as lightweight TFLite models for real-time inference on embedded systems.

- **Eye State Model:**
 - **Input:** Grayscale image of eye region, resized to 128×128 pixels.
 - **Output:** Single probabilistic value indicating whether the eye is open.
- **Yawn Detection Model:**
 - **Input:** RGB image of mouth region, resized to 96×96 pixels.
 - **Output:** Probability score indicating presence of a yawn.

6.2 Landmark-Based Analysis

Real-time facial landmark detection provides geometric features to complement CNN inference:

- **Eye Aspect Ratio (EAR):** Computed from vertical and horizontal eye landmarks; low EAR indicates eye closure.
- **Mouth Aspect Ratio (MAR):** Ratio of mouth height to width; high MAR suggests yawning.
- **Head Tilt Estimation:** Calculated using eye corner positions to detect nodding.
- **Occlusion Detection:** A separate hand landmark model is used to detect occlusions. If hand points are near the mouth’s center, potential yawns are invalidated to reduce false positives.

6.3 Hybrid Logic

The hybrid logic engine fuses outputs from CNNs and geometric analysis to ensure reliable detection:

- **Eye State Verification:** Eye closure is confirmed if either (a) EAR drops below a threshold or (b) CNN model predicts "closed" with high confidence.
- **Yawn Verification:** A yawn is detected if either (a) CNN score remains above threshold for multiple frames or (b) MAR exceeds a duration-based threshold.
- **Temporal State Buffering:** A sliding buffer (e.g., deque of 15 frames) ensures only persistent states trigger alerts, filtering out transient noise.
- **Cooldown Management:** Yawn alerts use cooldown timers to prevent multiple triggers from a single prolonged event.

7 Dataset and Training

The training and validation of the deep learning models were performed using a comprehensive aggregation of public and custom-built datasets. This section outlines the data sources, preprocessing pipeline, and training configuration.

7.1 Dataset Sources

The models were trained on extensive datasets to ensure robustness and high accuracy:

- **Eye State Dataset:** The eye state classification model was trained on the MRL Eye Dataset, a large-scale, publicly available collection of eye images.
- **Yawn Dataset:** The yawn detection model utilized a large, publicly available yawn dataset.
- **Dataset Scale:** Both datasets contained approximately 500,000 images, providing a diverse set of training examples.
- **Custom Data Supplement:** Additional data was captured using a standard webcam to incorporate variations in lighting, facial expressions, and head poses. This augmented the public datasets for real-world applicability.

7.2 Preprocessing Steps

A standardized preprocessing pipeline was applied before training the models:

- **Image Resizing:** Eye images were resized to 128×128 pixels and mouth images to 96×96 pixels to match their respective model input layers.
- **Color Space Conversion:** Eye images were converted to grayscale to emphasize textural features. Mouth images were retained in RGB format.
- **Pixel Normalization:** For improved training stability and easier convergence, all pixel values were normalized to the continuous interval $[0,1]$ by scaling each one with division by 255.0.

7.3 Training Details

The deep learning models were compiled and trained with standard configurations:

- **Loss Function:** Binary Cross-Entropy was used for both eye state and yawn classification tasks.
- **Optimizer:** The Adam optimizer was used for adaptive learning rate control and efficient convergence.
- **Training Procedure:** Each model was trained for 32 epochs using a 90:10 split for training and validation data.
- **Performance Metrics:** Model assessment while training was done by employing four performance metrics—Accuracy, Precision, Recall, and F1-Score that offered a comprehensive evaluation of both overall correctness of classification as well as balance between sensitivity and specificity.

7.4 Data Augmentation

To enhance model generalization and avoid overfitting, the following augmentation techniques were applied:

- Horizontal flipping
- Random brightness adjustments
- Addition of random noise

8 Experimental Results

This section presents both quantitative and qualitative evaluations of the proposed driver fatigue detection system. The analysis is structured to highlight the standalone performance of the CNN models, the advantages of the hybrid detection architecture, and the system’s runtime behavior on embedded hardware platforms.

8.1 Quantitative Comparison of CNN Models

The convolutional neural network models trained for eye state detection and yawning detection were tested with a special test dataset, and the confusion matrices yielded by them give a clear idea of their outstanding performance. The eye detection model is very precise, accurately detecting 41,333 "Close-Eyes" and 42,152 "Open-Eyes." The rates of misclassification are minimal, only 613 and 800 misclassifications each for each respective class, producing a very high overall accuracy of 98.34%. Such high precision shows the strength of the model and very high generalization power when classifying open and closed eye states. Similarly, the yawn detection model proves to be highly effective. It correctly identified 2,572 "no yawn" and 2,385 "yawn" samples. The very low false positive rate (19) and false negative rate (143) provides an incredible accuracy of 96.84%. This confirms the model can correctly discern oral movements from natural yawns, thus ensuring the global consistency of the system.

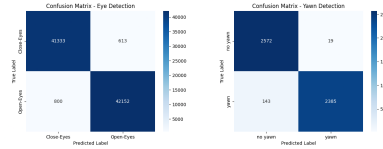


Fig. 2. Confusion matrix for developed models

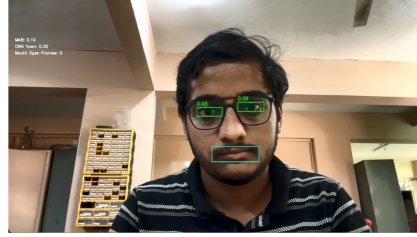


Fig. 3. Results of test setup

8.2 Comparative Analysis: Geometric vs. Hybrid Model

A comparative evaluation was conducted between the baseline “Geometric-Only” model and the proposed “Hybrid Model”. The geometric model relied solely on threshold-based EAR and MAR metrics, whereas the hybrid model incorporated both geometric and CNN-based predictions. **Geometric-Only Model:** While this approach was computationally light, it was prone to false positives due to head pose variation and inconsistent lighting. Such environmental variability occasionally caused erroneous EAR or MAR values, leading to inaccurate fatigue detection. **Hybrid Model:** The proposed hybrid model significantly reduced false positives through a robust fusion of geometric and CNN-based analysis. By integrating temporal state buffering using a deque structure, the system avoided transient misclassifications and relied on temporal consistency. For example, drowsiness is confirmed only when the number of ‘eye-closed’ frames within a buffered window exceeds a defined threshold. This strategy distinguishes prolonged closure from natural blinking. **Hybrid Decision Logic:** A logical OR operation governs the fusion process. Eye closure is flagged when either the EAR metric drops below a threshold or the CNN predicts eye closure with high confidence. This combination enables rapid detection of short-term events and robust confirmation of persistent states. The hybrid model demonstrated improved F1-Scores due to increased precision (fewer false alarms) while retaining high recall (sensitivity).

8.3 Edge Hardware Benchmarking

The system was deployed and tested on an embedded hardware platform (Raspberry Pi 4B) to assess real-time performance. **Frames Per Second (FPS):** The

system sustained a real-time video throughput with stable FPS. This ensured smooth and continuous monitoring under live operating conditions. A stable output of 60 fps was achieved during testing under the experimental setup. **Resource Discharge:** Usage of CPU and memory was under the risk level, highlighting the system’s suitability for low-power embedded systems. The deployment of TFLite models was instrumental in achieving efficient inference and minimal computational overhead. It was observed during testing that 80% of the resources(RAM, Computational capacity) were utilised in the setup. **Operational Stability:** Extended operation tests confirmed consistent performance without latency spikes or resource bottlenecks, demonstrating the system’s readiness for practical deployment.

9 Future Directive

- The addition of other sensors like heart rate sensors or EEG signals is meant to improve the extent and consistency with which we can perceive fatigue driving under varied conditions.
- Individualized adaptive model building assists in adjusting the thresholds of fatigue detection according to the behavior and conditions of each driver. This improves the sensitivity of the system as well as minimizes false alarms.
- Drawing inspiration from Vision Transformer (ViT) architectures for their high capacity to learn how facial features are interconnected. This can improve fatigue detection systems to be more accurate and robust even in varied settings.

10 Conclusion

This work introduces a strong hybrid fatigue detection system that synergistically integrates geometric landmark-based analysis with lightweight convolutional neural network (CNN) architectures to provide efficient and accurate real-time driver monitoring. Through visual feature classification coupled with geometric measurements obtained from facial landmarks, the system attains an optimal trade-off between detection accuracy and computational complexity, rendering it highly suitable for execution on low-resource edge devices. Large-scale experimental testing illustrates that the approach achieves a total classification accuracy of 98%, with stable real-time performance on the Raspberry Pi hardware platform. The results identify the presented method as a very encouraging solution for future-generation intelligent driver assistant systems, supporting improved road safety through accurate fatigue detection.

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